

1. Create a new GitHub repository.

Clone the repository to your local machine using SSH (generate an SSH key if needed, add the public key to your GitHub account).

Create a new branch named after your username

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps (main)
$ ssh-keygen -t rsa -b 4096 -C "pawartushar8485@gmail.com"
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/pawar/.ssh/id_rsa):
Enter passphrase for "/c/Users/pawar/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/pawar/.ssh/id_rsa
Your public key has been saved in /c/Users/pawar/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:m50HCeqne+6Z63rVGLs0aRSsqDF9Z1SH5+UGFB9+IhM pawartushar8485@gmail.com
The key's randomart image is:
+---[RSA 4096]-----+
|      .oE..      |
|      . ...o+..  |
|      = oo++ .   |
|      . . + o...ooo |
|      o o + S o= . |
|      + o + =+o.  |
|      . . B.o..   |
|      +o+..      |
|      +OBo.      |
+---+-----+

```

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps (main)
$ ssh-add ~/.ssh/id_rsa
Identity added: /c/Users/pawar/.ssh/id_rsa (pawartushar8485@gmail.com)

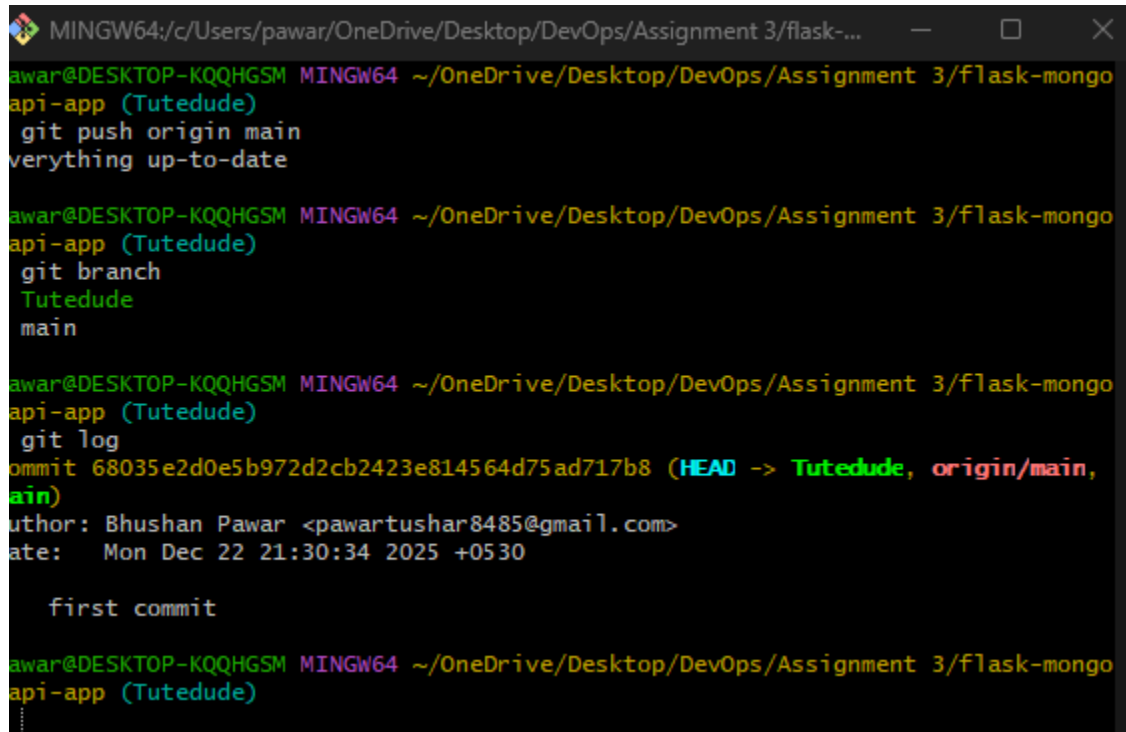
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps (main)
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACtAy33zo1NNGrdeU1HLki2aXepRxFiTRSkWpbPKhMx
U+szLXJkyoUFdjorj/UqKRhPuzkk0TqXm/1p4yO6vH92rzmZ8Co5IPa18R1jw015nHbEy5SbZ7kOX001
sRVQLRcXMuJUVE95P13lKmjCA6JFFrgT5f3oUZ4aK6I+yNHmyQqzPsH10u/kJyMZ5s90XPEB1kzTquI
CjdtiBL0vDfAuomDHKHfVgLyDnwsoeC2xH8g5H7mduwHJBWzT9Pt0dzxcv1mUmVw9Iv3kcu2XRI4Q47v
oTJHibZWzoEa5Cs7LQDfZP5YVY350q6h4wVXP1rV08HNkXx07IOFsCMjrh970bhNS2cwnGmt6rEmraIs
V5nYTxFc5e05JP23tGAQI52aPLM6XpCyNiKgU+cNrUXLhjJYp+/Oi azaGqe6xy2pP764wWGo7e2KZG7

```

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4 (main)
$ git clone https://github.com/tanuj-cmd-15/flask-mongo-api-app.git
Cloning into 'flask-mongo-api-app'...
remote: Enumerating objects: 19, done.
remote: Counting objects: 100% (19/19), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 19 (delta 0), reused 19 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (19/19), 76.22 KiB | 9.53 MiB/s, done.

```

- Create a new branch named after your username (e.g., Tutedude).
- Add your Flask project files to this branch.
- Commit the changes and merge the branch into the main branch.



```

MINGW64:/c/Users/pawar/OneDrive/Desktop/DevOps/Assignment 3/flask-mongo
awar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 3/flask-mongo
api-app (Tutedude)
git push origin main
Everything up-to-date

awar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 3/flask-mongo
api-app (Tutedude)
git branch
Tutedude
main

awar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 3/flask-mongo
api-app (Tutedude)
git log
commit 68035e2d0e5b972d2cb2423e814564d75ad717b8 (HEAD -> Tutedude, origin/main,
main)
author: Bhushan Pawar <pawartushar8485@gmail.com>
date: Mon Dec 22 21:30:34 2025 +0530

    first commit

awar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 3/flask-mongo
api-app (Tutedude)

```

2. Create a new branch named tanuj-cmd-15

- Update the content of the JSON file used for the /api route in this branch.
- Merge the tanuj-cmd-15 branch into the main branch.
- If there are conflicts during the merge, resolve them by accepting the changes from the tanuj-cmd-15 branch.
- Add the resolved changes to the staging area, commit them, and push the updates to the remote repository

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (Tutedude)
$ git checkout -b Tanuj-cmd-15
Switched to a new branch 'Tanuj-cmd-15'

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (Tanuj-cmd-15)
$ git add backend/backend_data.json

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (Tanuj-cmd-15)
$ git status
On branch Tanuj-cmd-15
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   backend/backend_data.json
```

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (Tanuj-cmd-15)
$ git commit -m "Update API JSON data"
[Tanuj-cmd-15 bf418be] Update API JSON data
 1 file changed, 2 insertions(+), 3 deletions(-)

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (Tanuj-cmd-15)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git merge Tanuj-cmd-15
Updating 68035e2..bf418be
Fast-forward
  backend/backend_data.json | 5 ++---
  1 file changed, 2 insertions(+), 3 deletions(-)
```

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 439 bytes | 439.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/tanuj-cmd-15/flask-mongo-api-app.git
  68035e2..bf418be  main -> main
```

3. Branch Creation:

- Create two branches: master_1 and master_2 from the main branch.
- **Feature Development in master_1:**
- In the master_1 branch, create a **To-Do Page** in the frontend.
 - The page should contain a form with the following fields:
 - **Item Name**
 - **Item Description**
- **Backend API in master_2:**
- In the master_2 branch, create a backend route named /submittodoitem.
- This route will:
 - Accept itemName and itemDescription via a POST request.
 - Store these details in a MongoDB database.
- **Merging Changes:**

Merge the changes from both master_1 and master_2 into the main branch.

```
$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git checkout -b master_1
Switched to a new branch 'master_1'

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (master_1)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git checkout -b master_2
Switched to a new branch 'master_2'

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (master_2)
$ |
```

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main|MERGING)
$ git checkout main
A       backend/api.py
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 656 bytes | 131.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tanuj-cmd-15/flask-mongo-api-app.git
   bf418be..22a45f6  main -> main

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$
```

Enhancing the To-Do Form in master_1:

- In the master_1 branch, add the following fields to the To-Do form:
 - **Item ID**
 - **Item UUID**
 - **Item Hash**
- **Committing in Sequence:**
- Add and commit each field separately in the following order:
 - **First commit:** Add **Item ID** field.
 - **Second commit:** Add **Item UUID** field.
 - **Third commit:** Add **Item Hash** field.

```

api-app (master_1)
$ git add .

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (master_1)
$ git commit -m "Add Item ID field"
[master_1 350f4aa] Add Item ID field
1 file changed, 1 insertion(+)

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (master_1)
$ git add .

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (master_1)
$ git commit -m "Add Item UUID filed"
[master_1 716abe5] Add Item UUID filed
1 file changed, 1 insertion(+)

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (master_1)
$ git add .

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (master_1)
$ git commit -m "ADD Item Hash field"
[master_1 9f02e7d] ADD Item Hash field
1 file changed, 1 insertion(+)

```

- **Merging to main:**
- Merge the master_1 branch into the main branch.

```

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (master_1)
$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (main)
$ git merge master_1
Updating 55910b2..9f02e7d
Fast-forward
 frontend/templates/todo.html | 3 +++
 1 file changed, 3 insertions(+)

```

```

1 file changed, 3 insertions(+)

pawar@DESKTOP-KQQHGSM MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-
api-app (main)
$ git log --oneline
9f02e7d (HEAD -> main, master_1) ADD Item Hash field
716abe5 Add Item UUID field
350f4aa Add Item ID field
55910b2 erase all
1aeafc5 (origin/main, origin/HEAD) Add item Hash field
fe96246 Add item UUID field
226eeb4 Add item id field
22a45f6 Add To-Do form page
bf418be (Tanuj-cmd-15) Update API JSON data
68035e2 (list, Tutedude) first commit

```

- **Git Reset and Commit Deletion:**

- In the main branch, use **Git Reset** to roll back to the commit where only the **Item ID** field was added.
- Use `git reset --soft` to ensure changes remain staged.
- Re-commit this state to the main branch.
- Merge this updated state to the main branch.

- **Rebasing Changes:**

- Rebase the updated changes in the main branch to the master_1 branch.

Clarification:

- During rebasing, **preserve individual commits** to maintain the commit history for each change (i.e., do not squash commits).
- Use `git rebase main master_1` to integrate changes from the main branch back into the master_1 branch.

```
pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git reset --soft 9f02e7d

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git commit -m "revert to Item ID only"
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git push origin main --force
Enumerating objects: 24, done.
Counting objects: 100% (24/24), done.
Delta compression using up to 8 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (20/20), 1.78 KiB | 364.00 KiB/s, done.
Total 20 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), completed with 2 local objects.
To https://github.com/tanuj-cmd-15/flask-mongo-api-app.git
   1aeafc5..9f02e7d  main -> main

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (main)
$ git checkout master_1
Switched to branch 'master_1'

pawar@DESKTOP-KQQHGS MINGW64 ~/OneDrive/Desktop/DevOps/Assignment 4/flask-mongo-api-app (master_1)
$ git rebase main
Current branch master_1 is up to date.
```